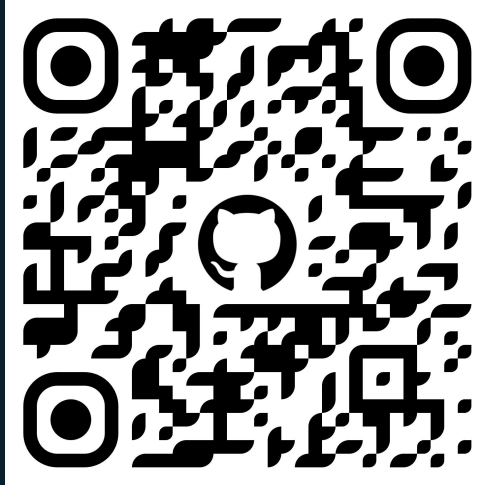


# **“Programming”**

## **<Recap>**

# Resource



[github.com/ro-witthawin/TBS-BI303-2026](https://github.com/ro-witthawin/TBS-BI303-2026)

## TBS-BI303-2026



## Class schedule

| Date       | Topic                         | Material              |
|------------|-------------------------------|-----------------------|
| 2026-08-03 | Intro to Programming Language | <a href="#">Day01</a> |
| 2026-08-10 | Basic String & List           | <a href="#">Day02</a> |
| 2026-08-17 | Function                      | <a href="#">Day03</a> |
| 2026-08-24 | Loop                          | <a href="#">Day04</a> |
| 2026-08-31 | Dictionary                    | <a href="#">Day05</a> |
| 2026-09-07 | Class Object                  | <a href="#">Day06</a> |
| 2026-09-14 | Freq Library                  | <a href="#">Day07</a> |
| 2026-09-21 | Final Exam                    |                       |

# Grading

A :  $\geq 80$

B+ : 75-79

B : 70-74

C+ : 65-69

C : 60-64

D+ : 55-59

D : 50-54

F :  $< 50$

[2026-08-31] Exam #1 : 40%

- String & List
- Function
- Loop

[2026-09-21] Exam #2 : 60%

- String & List
- Function
- Loop
- Dict
- Class

# Logical Thinking

$$Q : 1 + 2 + (3 + 4) = ?$$

# Programing Step

```
x = 15

if x > 10:
    print("x is greater than 10")
elif x == 10:
    print("x is exactly 10")
else:
    print("x is less than 10")
```

# Basic Variable

## Variable Naming rules

x = "I" 

X0 = "love" 

x\_1 = "program" 

2x = "ing" ✗

x = True # Boolean

x = False # Boolean

x = 10 # Integer

x = 10.0 # Float

x = "Hello" # String

# Basic Command & Operation

```
print(".....")  
print('.....')
```

```
type(...) # Check data type  
bool(...) # Change data to Boolean  
int(...) # Change data to Integer  
float(...) # Change data to Float
```

```
x = 10  
print(x)  
# Note & Comment  
print("Hello") # Note
```

# Basic Operation

$x = 1 > 2$  # Greater than

$x = 2 < 3$  # Less than

$x = 3 == 4$  # Equal to

$x = 4.4 != 5$  # Not equal to

$x = 5 >= 6$  # Greater or equal

$x = 6 <= 7$  # Less or equal

$x = 1 + 2$  # Addition

$x = 2 - 3$  # Subtraction

$x = 3 * 4$  # Multiplication

$x = 4 / 5$  # Division

$x = 5 // 6$  # Floor Division

$x = 6 \% 7$  # Modulus

$x = 7 ** 8$  # Exponentiation

$x += 10$  # Short of  $x = x + 10$

$x -= 10$  # Short of  $x = x - 10$

$x *= 10$  # Short of  $x = x * 10$

$x /= 10$  # Short of  $x = x / 10$

# And, Or, Not

x = True and True

x = False or True

x = not True

X = not(False)

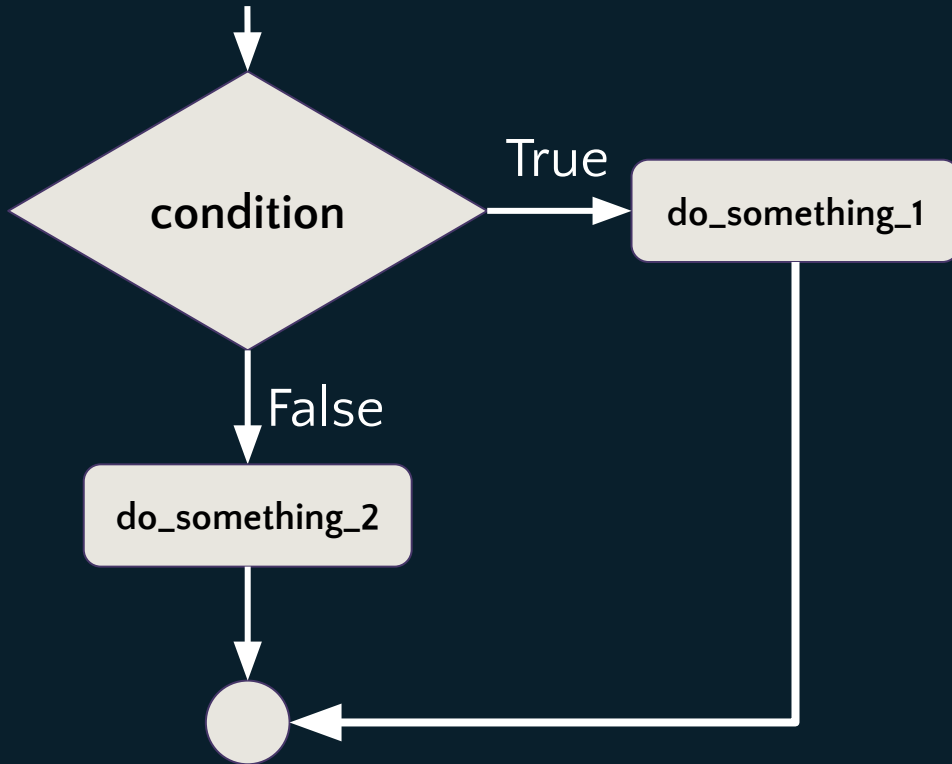
| X     | Y     | X & Y |
|-------|-------|-------|
| True  | True  | True  |
| True  | False | False |
| False | True  | False |
| False | False | False |

And

| X     | Y     | X   Y |
|-------|-------|-------|
| True  | True  | True  |
| True  | False | True  |
| False | True  | True  |
| False | False | False |

Or

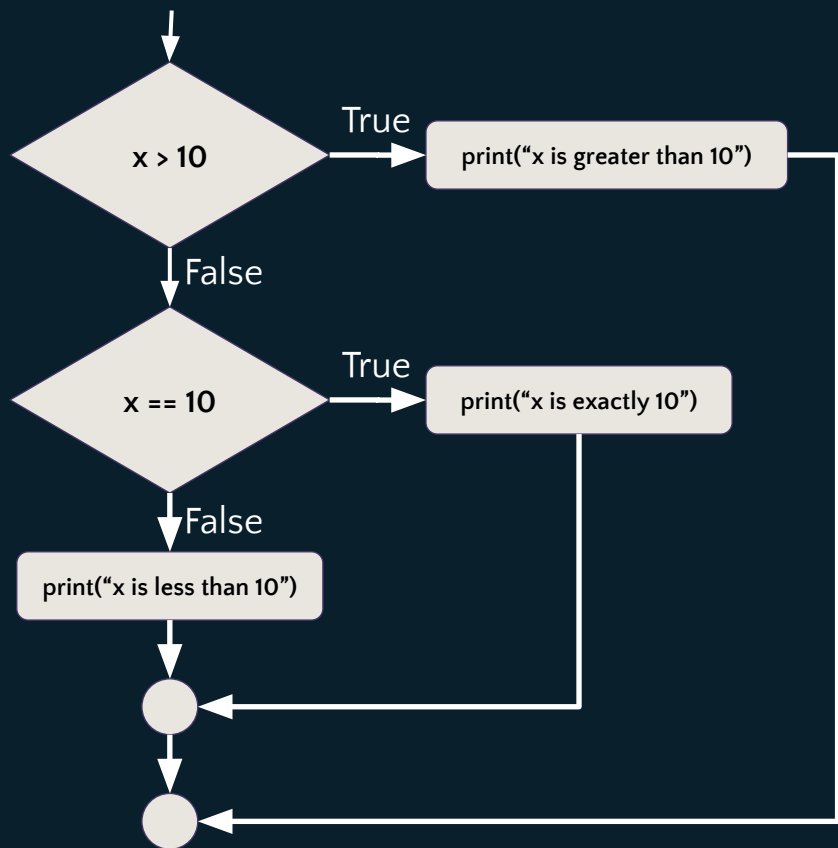
# If / Else



```
x = 15
```

```
if x > 10:  
    print("x > 10")  
else:  
    print("x <= 10")
```

# If / Else



```
x = 15
```

```
if x > 10:  
    print("x is greater than 10")  
else:  
    if x == 10:  
        print("x is exactly 10")  
    else:  
        print("x is less than 10")
```

```
x = 15
```

```
if x > 10:  
    print("x is greater than 10")  
elif x == 10:  
    print("x is exactly 10")  
else:  
    print("x is less than 10")
```

# Condition with And/Or

x = True and True

x = False or True

x = not True

X = not(False)

# Grade Calculation

score = 70

if (x >= 80) and (x <= 100):

    print("A")

elif x >= 70 and x < 80:

    print("B")

else:

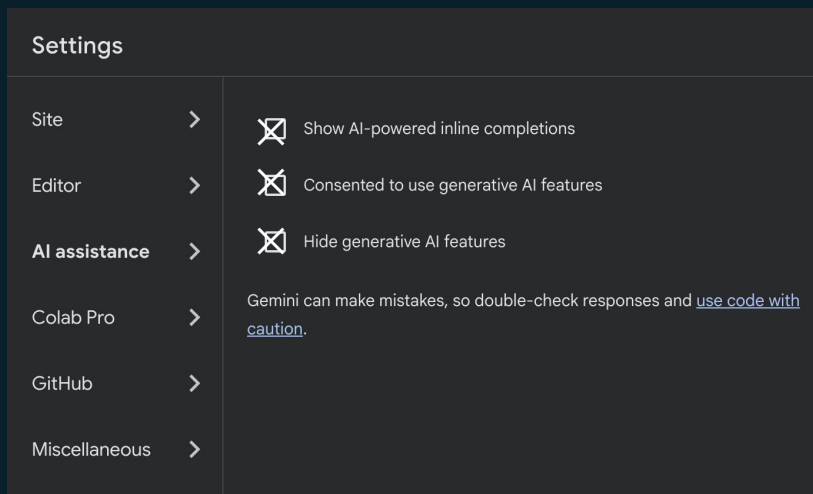
    print("Other")

# Hand-on 15 min without AI



## Credit Approval

- ถ้า Credit Score  $\geq 700$  และ รายได้ต่อเดือน  $\geq 50,000$  ให้อนุมัติทันที
- ถ้า Credit Score  $< 600$  หรือ รายได้ต่อเดือน  $< 20,000$  ให้ปฏิเสธ
- นอกนั้นส่งให้เจ้าหน้าที่ตรวจสอบเพิ่มเติม



Input : credit\_score, income

Output : status # ["approve", "reject",  
"wait for review"]

**“String”**

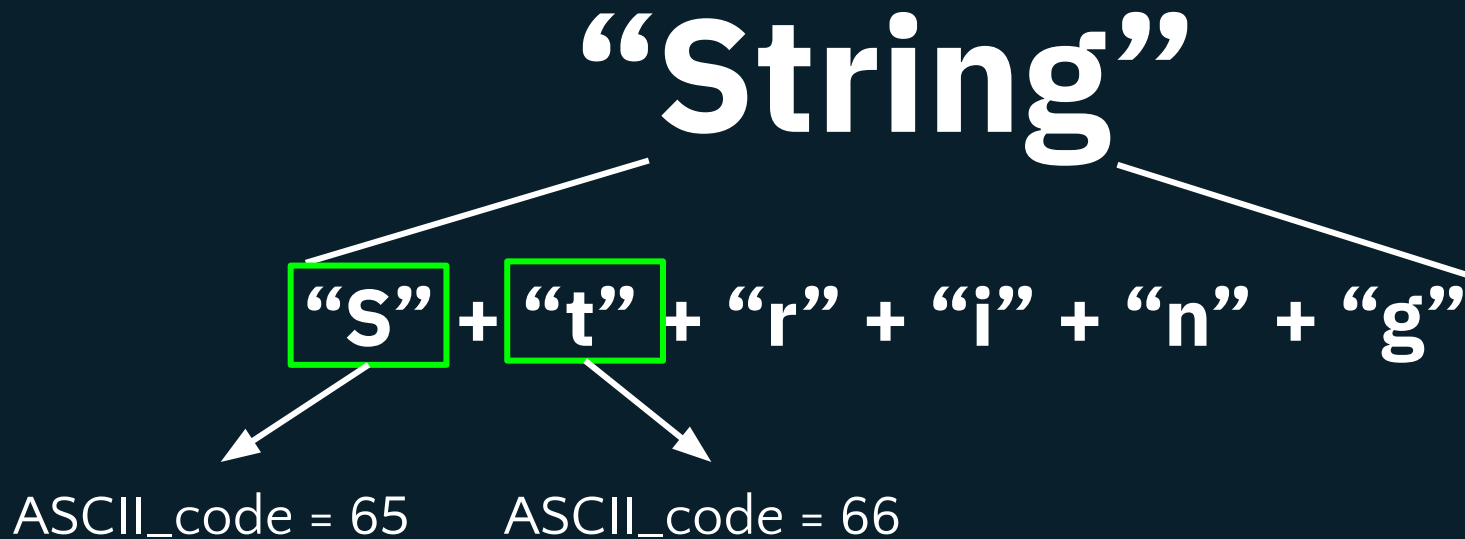
# String

```
print(".....")  
print('.....')
```

**“String”**

“S” + “t” + “r” + “i” + “n” + “g”

# String



# String

```
print(".....")  
print('.....')
```

**“String”**

**“S” + “t” + “r” + “i” + “n” + “g”**

Index

[0]

[1]

[2]

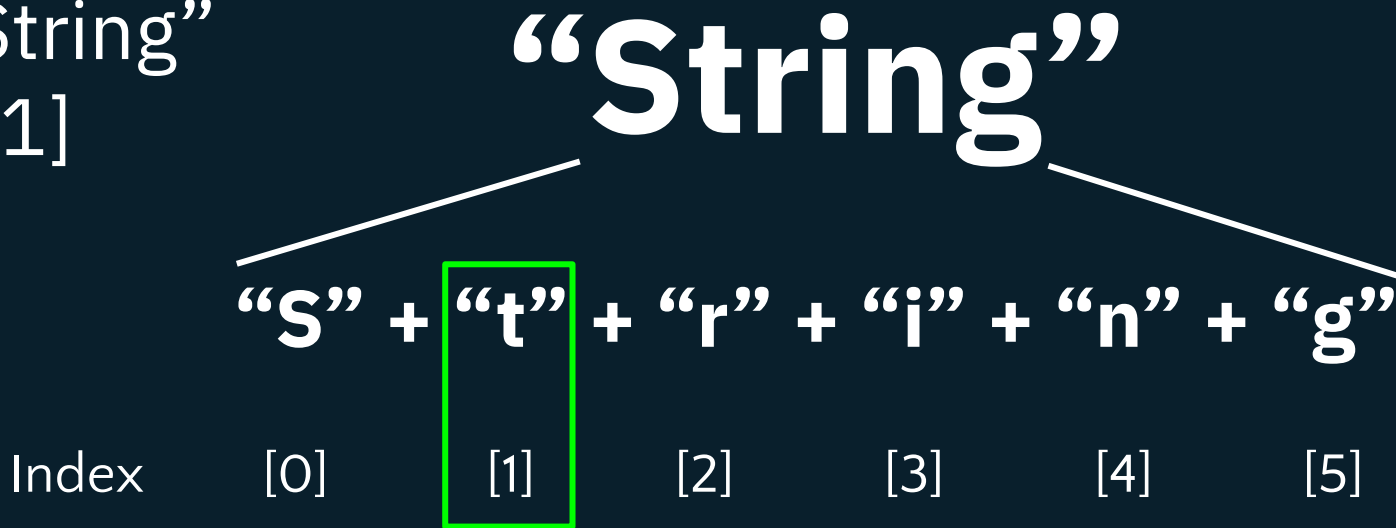
[3]

[4]

[5]

# String

```
a = "String"  
b = a[1]
```



```
b = "t"
```

# String

```
a = "String"  
b = a[0: 3]
```

**"String"**

**"S" + "t" + "r" + "i" + "n" + "g"**

Index

[0]

[1]

[2]

[3]

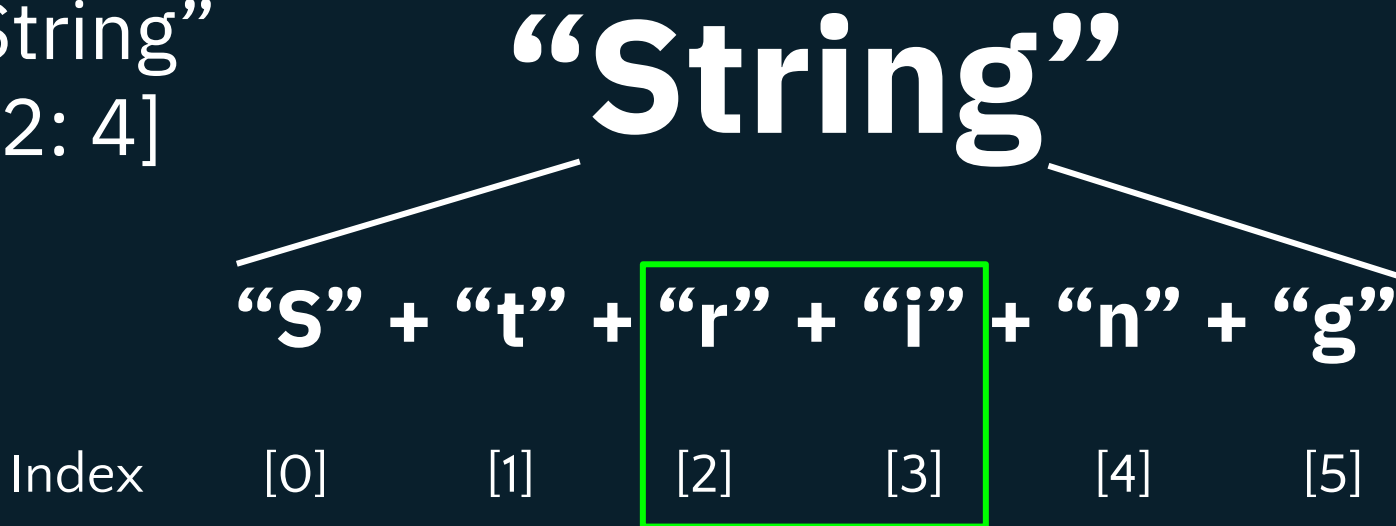
[4]

[5]

```
b = "Str"
```

# String

```
a = "String"  
b = a[2: 4]
```



```
b = "ri"
```

# String

```
a = "String"  
b = a[4:]
```

**“String”**

**“S” + “t” + “r” + “i” + “n” + “g”**

Index

[0]

[1]

[2]

[3]

[4]

[5]

```
b = “ng”
```

# String

```
a = "String"  
b = a[:4]
```

**"String"**

**"S" + "t" + "r" + "i" + "n" + "g"**

Index

[0]

[1]

[2]

[3]

[4]

[5]

```
b = "Stri"
```



# String

**“String”**

**“S” + “t” + “r” + “i” + “n” + “g”**

|       |     |     |     |     |     |     |
|-------|-----|-----|-----|-----|-----|-----|
| Index | [0] | [1] | [2] | [3] | [4] | [5] |
|-------|-----|-----|-----|-----|-----|-----|

|                |      |      |      |      |      |      |
|----------------|------|------|------|------|------|------|
| Negative Index | [-6] | [-5] | [-4] | [-3] | [-2] | [-1] |
|----------------|------|------|------|------|------|------|

# String

**“String”**

a = “String”

b = a[-3:]

**“s” + “t” + “r” + “i” + “n” + “g”**

Index

[0]

[1]

[2]

[3]

[4]

[5]

Negative Index

[-6]

[-5]

[-4]

[-3]

[-2]

[-1]

# String: Indexing

# “String”

a = “String”

b = a[-3:]

“s” + “t” + “r” + “i” + “n” + “g”

Index

[0]

[1]

[2]

[3]

[4]

[5]

Negative Index

[-6]

[-5]

[-4]

[-3]

[-2]

[-1]

b = “ing”

# String: Indexing

# “String”

a = “String”

b = a[-3:-2]

“s” + “t” + “r” + “i” + “n” + “g”

Index

[0]

[1]

[2]

[3]

[4]

[5]

Negative Index

[-6]

[-5]

[-4]

[-3]

[-2]

[-1]

b = “i”

# String: Indexing

# “String”

a = “String”

b = a[-3:-5]

“s” + “t” + “r” + “i” + “n” + “g”

|       |     |     |     |     |     |     |
|-------|-----|-----|-----|-----|-----|-----|
| Index | [0] | [1] | [2] | [3] | [4] | [5] |
|-------|-----|-----|-----|-----|-----|-----|

|                |      |      |      |      |      |      |
|----------------|------|------|------|------|------|------|
| Negative Index | [-6] | [-5] | [-4] | [-3] | [-2] | [-1] |
|----------------|------|------|------|------|------|------|

b = “”

# String: Operation

`"Hello Python".upper()` # "HELLO PYTHON"

`"Hello Python".lower()` # "hello python"

`" Hello Python ".strip()` # "Hello Python"

`"Hello Python".replace("Python", "World")` # Hello World

`"Hello Python".count("l")` # 2

`"Hello Python".find("o")` # 4

`str(3.14)` # "3.14"

**“List”**

**List**

**[0, 1, 2, “a”, “B”]**

# List

**[0, 1, 2, “a”, “B”]**

|       |     |     |     |     |     |
|-------|-----|-----|-----|-----|-----|
| Index | [0] | [1] | [2] | [3] | [4] |
|-------|-----|-----|-----|-----|-----|

|                   |      |      |      |      |      |
|-------------------|------|------|------|------|------|
| Negative<br>Index | [-5] | [-4] | [-3] | [-2] | [-1] |
|-------------------|------|------|------|------|------|